

Swoop: Animating Characters Along a Path

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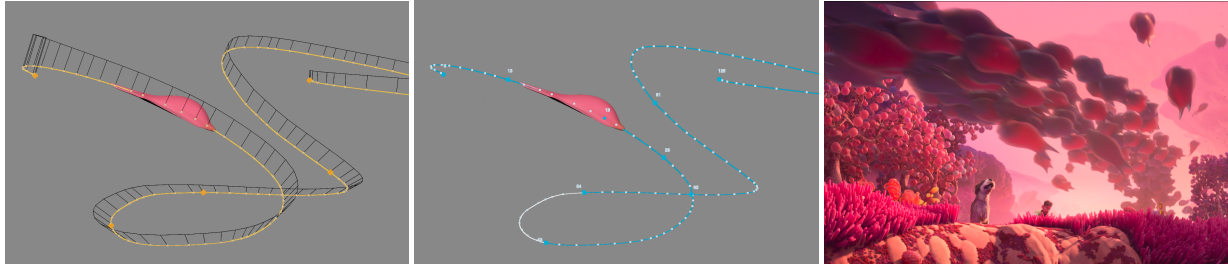


Figure 1: Animating Creatures along a Path: Shape Mode, Spacing Mode, and Final Render ©2023 Disney

ABSTRACT

Swoop is an in-house Disney tool that allows artists to easily animate motion along a trajectory. It provides an intuitive interface for users to author and edit a Swoop curve, bind one or more objects to that Swoop curve, and easily animate the timing, spacing, twist, and offset of the bound objects.

KEYWORDS

procedural animation, crowd animation, motion path manipulation

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1 INTRODUCTION

Walt Disney Animation Studios has long had many robust systems for rigging and animating skeletal characters, as well as directing motion along a surface path. However, the toolset for animating more fluid or abstract motion paths through complex 3D space was, until recently, much more limited. When confronted with this challenge for Gale, an incorporeal wind spirit character in “Frozen 2”, we developed Swoop, a specialized curve-based animation tool.

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Swoop provides an intuitive, sketch-based interface for users to author and edit a dense Swoop curve. This curve defines the trajectory of motion and is used as a tactile mechanism for adjusting the timing of objects moving along the curve. With the additional creative challenges of “Strange World”, the tool was expanded to allow for multiple bindings per Swoop curve, controls for inter-object spacing, offset buffers, and animatable twist controls along the curve. These features of the Swoop workflow have enabled more complex and creative character animation at the studio.

2 APPROACH

The trajectory of the Swoop curve is represented by a custom Maya shape node called MultiCurve, which was originally developed for hair. The tool offers two editing modes, illustrated in Figure 1. Shape mode, indicated by an orange curve, edits the 3D points along the trajectory. Spacing mode, indicated by a blue curve, edits the time at which the object reaches each point. Both modes allow artists to add anchor points along the curve, breaking it into segments that will either be influenced by or isolated from edits. Figure 2 shows the node graph configuration of the Swoop curve.

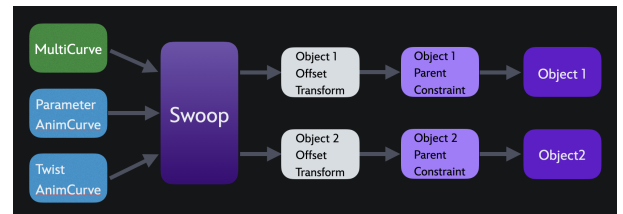


Figure 2: Node Graph for Swoop ©2023 Disney

2.1 Shape Manipulation

The shape of the path can be edited either by grabbing a point on the curve with the mouse and dragging it, or sketching a curve along a plane parallel to the camera plane. The curve is rebuilt with each movement and its parameterization is updated in order to maintain a consistent curve density while allowing the artists to drastically change its length and shape.

2.2 Spacing/Timing Manipulation

The spacing of an object traveling along a curve is visualized as small dots representing the location of the object at each frame. Artists can click and drag on the dots to slide them along the curve within a segment, changing the speed at which the object travels that portion of the path. In order to make the editing feel natural, we allow the mouse drag to drift far away from the curve while still maintaining reasonable and smooth edits. We also redistribute the affected frames using an algorithm that ensures the transition between the spacing on either side of the selected frame remains smooth and natural. The curve can also be retimed in spacing mode. This operation changes the number of frames it takes to move along the path while still maintaining the overall spacing.

2.3 Brush Workflow

Each mode has a version of two brushes: smooth and nudge/smudge. In shape mode, the smooth brush averages out the 3D locations of points using a weighted average of points within the brush radius. The smudge/nudge brush pushes points in the direction of the brush stroke. Both work in similar ways with spacing, but instead affect the frames along the path instead of the shape itself.

2.4 Multiple Bound Objects

The top level transform node of an object can be bound to a Swoop curve to drive the overall transformation of the object. Multiple controls, such as spine IK handles of a blood cell creature in “Strange World,” can be bound to a Swoop curve to generate a sliding motion along the curve. For each bound object, an attribute is used to specify an offset distance from the adjacent bound object along the curve. This offset distance is calculated automatically at the moment of binding. Users can apply an offset distance multiplier attribute to control and animate the distance between bound objects, making it easy to animate stretch and squash.

2.5 Rotation/Translation with Twist

The underlying MultiCurve defines a normal at any given point, by parallel transporting a base normal vector down the curve. We construct a unique local frame from the normal and tangent vectors of a point. This local frame defined by the MultiCurve has its Z-axis aligned with the tangent vector and Y-axis aligned with the normal vector. To provide an additional control for twisting the normal vector about the tangent vector, a twist angle attribute is added, as shown in Figure 2. A final reference frame for bound objects is then constructed using the tangent vector and the twisted normal vector.

Any changes to the shape of the MultiCurve in an area will affect the normal vectors and, therefore, the reference frames for the bound objects down the curve. Since it is undesirable for a local

shape manipulation to have a widespread effect on the orientation of bound objects down the curve, we introduce an automatic function to adjust the twist angles to compensate for the changes of normal vectors on the MultiCurve. The twist compensation is applied for changes of spacing and timing as well in order to maintain the orientation of bound objects at any point on the curve.

2.6 Offset Transform Buffers

When there are multiple characters bound to a Swoop curve, there is a need to animate some bound objects away from the curve to add variation to the animation. An offset transform node is inserted between the reference frame generated by the Swoop node and the parent constraint (see Figure 2) of the bound object. The offset transform node can be animated to create an offset animation from the reference frame. The combined transformation of the reference frame and the offset transform node is applied to the bound object through the parent constraint node.

2.7 Baking and Unbaking

Animators can bake out Swoop animation to keyframe animation for further tweaking. This operation detaches the bound objects from the Swoop curve. If animators want to modify the trajectory of an object, they can reconstruct a Swoop curve (i.e. unbake) from the keyframe animation of the object. This baking and unbaking process makes it very easy for animators to modify the trajectory, timing, and spacing of a character moving along a path.

3 RESULTS

In “Strange World,” Swoop was used in many shots. One, shown in Figure 3, used a single Swoop curve to animate hundreds of blood cell creatures moving along a path. Each object or character along a path can have its own timing and offset, making it appear individually animated. Swoop makes it easy for animators to use these additions to fine tune the performance of each individual along the path, as well as make large scale changes to the scene by operating on the Swoop curve.

4 CONCLUSIONS AND FUTURE WORK

Swoop allowed our artists to author and edit complex motion paths quickly and intuitively. It has evolved through artist feedback and production needs and made it possible for our animators to more easily iterate on their characters’ performance, improving the artists’ experience. Currently Swoop is implemented in Maya, but as our studio is exploring a wider range of DCCs, we plan to port it for future use.

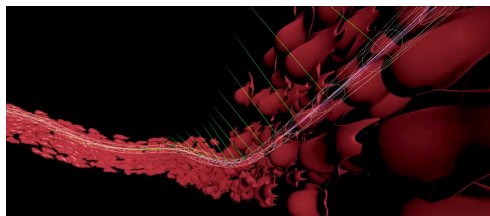


Figure 3: Usage of Swoop in “Strange World” ©2023 Disney